

VALERIO DIONISI

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CURRENT POSITIONS

» Doctoral student in Economics and Statistics, *University of Milano-Bicocca* *Milan, Italy*

RESEARCH INTERESTS

Macroeconomics, Production Networks, Structural Macro-Labour, Fiscal and Monetary Policy

EDUCATION

2024 – 2025	Visiting Ph.D. Student UNIVERSITY OF OXFORD <i>Activities:</i> Production networks, macro-labour, fiscal policy <i>Advisor:</i> Prof. Francesco Zanetti
2023 – 2024	Visiting Ph.D. Student UNIVERSITAT POMPEU FABRA <i>Activities:</i> Numerical techniques for quantitative macroeconomics, production networks <i>Advisor:</i> Prof. Davide Debortoli
2021 – present	Ph.D. in Economics, Statistics and Data Science UNIVERSITY OF MILANO-BICOCCA <i>Topics:</i> Macroeconomics, production networks, macro-labour <i>Thesis:</i> “Macroeconomic Perspectives on Sectoral Composition: Networks and Inequality” <i>Supervisor:</i> Prof. Andrea Colciago
2019 – 2021	Ms.C. in European Economy and Business Law (summa cum laude) TOR VERGATA UNIVERSITY <i>Thesis:</i> “Evidence of Debt Consolidation Dynamics for Europe and Italy” <i>Supervisor:</i> Prof. Barbara Annicchiarico , <i>Co-supervisor:</i> Prof. Leonardo Becchetti
2016 – 2019	B.A. in Political Sciences ROMA TRE UNIVERSITY <i>Thesis:</i> “Instrument-Target Alignment and the Coordination of Economic Policies” <i>Supervisor:</i> Prof. Gian Cesare Romagnoli

FURTHER EDUCATION

2022 *Advanced Computational Macroeconomics* LONDON SCHOOL OF ECONOMICS

RESEARCH PAPERS

Job Market Paper ***The Horizontal Geometry of Production Networks*** [GO TO THE PAPER](#)

Modern economies are organized as dense production networks, yet sectoral comovement is typically understood only through vertical Input-Output linkages and cascade effects. This paper argues that this view is incomplete. I show that comovement fundamentally depends on horizontal complementarities: the extent to which sectors share common upstream suppliers or downstream buyers. These similar demand and supply network structures induce simultaneous responses to idiosyncratic shocks, shaping the sign of comovement. I develop a theoretical framework that disentangles vertical from horizontal transmission, offering a unified view of comovement rooted in the horizontal geometry of common economic interactions rather than their trade intensity. Unlike vertical propagation, horizontal transmission does not rely on cascading amplification or dominant sectors for aggregate fluctuations, but generates systematic comovement that prevents shocks from averaging out. Using U.S. Input-Output data, I show that sectoral employment responses vary systematically with demand and supply distances, with nearby sectors exhibiting dampened or opposing responses and distant sectors comoving positively.

Industry dimension is increasingly dominant to investigate the upward trend of inequality. This paper examines the key drivers of U.S. wage inequality, emphasising the role of heterogeneous capital-labour substitution elasticities across industries in shaping wage dispersion. Key is the distinction of a *quantity effect* (changes in the composition of capital and labour inputs) and a *structural effect* (reflecting technological transformations in inputs substitutability) from Skill-Biased Technological Change (SBTC). Findings suggest that industry-level transformations on the labour side – differentials in job tasks substitutability and workforce composition – constitute the principal drivers of real wage inequality, overshadowing the contribution of capital-side adjustments. A structural estimation of the model reveals that trend-asymmetries in the elasticities of substitution between ICT capital, routine and non-routine workers account for 94% of observed wage variance, while stronger sorting and segregation effects further exacerbate such dispersion. Upon neutralising structural differences between industries, quantity adjustments reckons merely 6-15% of the observed wage inequality.

with Colciago A. (DNB, UniMiB), Siena D. (PoliMi, Bocconi), and Zago R. (BdF, CdF, ESCP)

with Bertucci A. (UniPv)

with Siena D. (PoliMi, Bocconi)

TEACHING

2024/2025-still	International Macroeconomics (Grad) , TA Lecturer	POLYTECHNIC UNIVERSITY OF MILAN
2024/2025-still	Macroeconomics (UndGrad) , TA Lecturer	POLYTECHNIC UNIVERSITY OF MILAN
2023/2024-still	Computational Macroeconomics (Grad) , TA Lecturer	UNIVERSITY OF MILANO-BICOCCA
2023/2024-still	Dynamic Asset Pricing (Grad) , TA	UNIVERSITY OF MILANO-BICOCCA
2023/2024-still	Economia dei Contratti Finanziari (Grad) , TA	UNIVERSITY OF MILANO-BICOCCA
2023/2024-still	Macroeconomics (UndGrad) , TA	UNIVERSITY OF MILANO-BICOCCA

SKILLS

<i>Languages:</i>	ITALIAN (native), ENGLISH (fluent), FRENCH and SPANISH (intermediate)
<i>Programming:</i>	MATLAB, DYNARE, STATA (advanced); R (proficient); L ^A T _E X (advanced)
<i>Microsoft Office:</i>	WORD, EXCEL, POWERPOINT, OUTLOOK

CERTIFICATES

2021	Introduction to Linear Algebra	MATLAB
2021	Introduction to Statistical Methods	MATLAB
2021	Fundamentals	MATLAB
2021	OnRamp	MATLAB
2021	Optimization OnRamp	MATLAB

CONFERENCES AND SEMINARS

	3rd Milan Ph.D. Economics Workshop, <i>Organizer</i> ; University of Oxford, Naples School of Economics; 4th PhD and Post-Doctoral Workshop, Roma 3 University (<i>scheduled</i>), “Sapienza” University of Rome, Catholic University of Milan, University of Milano-Bicocca, <i>Presenter</i>
2025	2nd Milan Ph.D. Economics Workshop, <i>Organizer</i> , University of Milano-Bicocca, University of Pavia, <i>Presenter</i>
2024	

REFERENCES

ANDREA COLCIAGO

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FRANCESCO ZANETTI

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